Assignment 2 - LLM Document Analysis

CAP5619 | Methodology Report | November 4, 2026

GitHub repository: https://github.com/danielp382/Assignment-2---LLM-Document-AnalysisV2

Project objective and setup

For this assignment, I used Python to read SEC Form 8-K filings and exhibits and check whether a company announced a new product. I used Ollama with the local llama3.2:3b model instead of a paid LLM API. I started by working with the SEC documents in Google Colab, then moved the analysis to my Windows computer so I could run the model locally.

How I extracted the information

I collected filing text from SEC EDGAR and used BeautifulSoup to remove HTML formatting when needed. I gave LLaMA instructions to answer Yes or No, return the product name and a description under 180 characters, and give a short reason. The prompt tells the model not to count financial reporting changes, existing products, or unrelated company announcements as new products. In Python, I read the model response line by line, stored the fields in a dictionary, and checked for missing values and descriptions that were too long.

SEC data and CSV output

I used the SEC submissions API and the exact filing accession number to retrieve the company name, stock ticker, and filing acceptance time. I kept this information separate from the LLM response because the SEC API is the source for filing metadata. I saved the results with Python's csv library using the five required columns: company_name, stock_name, filing_time, new_product, and product_description. I also used a separate validation script to check the column names, populated fields, and description length.

Additional scripts and testing

After confirming the first two examples worked, I added collect_8k_candidates.py to search SEC filings for possible new-product announcements. I then ran assignment2_bulk.py with my local LLaMA model to screen the collected filings and save the positive results to product_announcements.csv. I used check_final_csv.py to check the required headers, missing values, and the 180-character limit. The batch script also saved a review_log.csv so I could see what happened with each filing.

Results from manual testing and batch processing

I first manually checked two filings from two companies. TELA Bio's August 16, 2023 exhibit announced OviTex PRS Long-Term Resorbable, and the model correctly answered Yes. Microsoft's October 28, 2026 filing discussed changes in financial segment reporting, so the model correctly answered No. For the additional batch, the collector gathered 116 candidate SEC documents. I completed 20 document analyses and stopped while document 21 was starting. LLaMA saved 7 possible product announcements to the CSV. My validation script confirmed five correct columns, seven populated records, and descriptions below 180 characters. These are automated results; I did not finish manually verifying all seven against the original filings.

Challenges and limitations

The biggest challenge was finding 8-K documents that actually announced new products instead of earnings, agreements, or other company updates. Early on I also had to fix a Windows text-encoding error and make sure I was using the correct SEC accession number. The model sometimes gave slightly different descriptions on repeated runs, so I added checks for missing fields and length. I also learned that a CSV row can pass formatting checks and still need a manual accuracy review.

Why my CSV has fewer than 100 entries

I initially misunderstood the 100-entry expectation and spent most of my time making sure the Python script, local model, SEC metadata lookup, and CSV output were working correctly. Once I realized the grading criteria expected 100 entries, I expanded the program to collect and process more filings. However, because of the submission deadline, I stopped after 20 completed batch documents rather than finish all 116 candidates. The CSV currently has 7 LLaMA-identified product records, not 100. I am reporting the actual results instead of repeating entries or making up product announcements.

What I would improve

With more time, I would finish processing the remaining candidate filings, manually verify each Yes result against its SEC exhibit, check whether similar names are duplicates or separate product launches, and update the CSV accordingly. I would also improve the script to capture more than one new product from the same filing. This assignment helped me understand that getting an answer from an LLM is only part of the process; the output still needs to be validated.